INDEX NUMBER..... YEAR.....

SECTION A

Questions 1 to 20: *True (T) or False (F)*

1. A centre line is a thin line. A: T B: F
2. A soft lead is recommended for drawing. A: T B: F
3. A Set-square is required for horizontal lines. A: T B: F
4. Technically, scale is defined as a ratio of Actual : Drawn A: T B: F
5. A regular curve cannot be drawn with a French Curve. A: T B: F
6. Constructional lines are thin lines. A: T B: F
7. The Path traced by a point attached on a circle that rolls on the outside of another circle is called inferior hypotrochoid. A: T B: F
8. The Path traced by a point attached inside a circle that rolls on a straight line is called inferior cycloid. A: T B: F
9. Epitrochoids, Involutes of polygons, Spirals and Helices are space curves A: T B: F
10. On the basis of Distance between Object and Ray, projections can be termed as perspective and parallel ray projections. A: T B: F
11. The symbol,  , signifies a First Angle Orthographic Projection A: T B: F
12. The distance between the centres of two internally tangential arcs is: $\overline{Cc} = R + r$ A: T B: F
13. Hidden outlines are indicated in drawings with thick continuous lines. A: T B: F
14. Only line type A (thick continuous) and line type D represent edges. A: T B: F
15. The lengths and spacing of the elements of thick chain lines should differ from those of thin chain lines. A: T B: F
16. When thin chain lines are used as centre lines, they should cross one another at short portions of the line. A: T B: F
17. All chain lines, E, should start and finish with a short dash. A: T B: F
18. Dashed lines should meet at corners and tangent points with dashes. A: T B: F
19. Where visible line and invisible line coincide in a drawing, the visible line takes precedence. A: T B: F
20. When invisible line and centre line coincide in a drawing, the centre line takes precedence. A: T B: F

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INDEX NUMBER..... YEAR.....

21. Outlines of visible edges can be represented by:
    A. Thin continuous lines
    B. Thick Short lines
    C. Thick continuous line
    D. Thick chain lines
    E. None of the above

22. The ratio of thick lines: thin line recommended for standard drawings is:
    A. 1:1
    B. 1:2
    C. 1:3
    D. 2:1
    E. 3:1

23. The following information cannot be found in the title block:
    A. Dimension of object
    B. Title of drawing
    C. Drawn Scale
    D. Names and dates of drafter, checker, designer
    E. Name of firm.

24. A scale of 1:2 represents:
    A. A magnification
    B. A reduction
    C. A full-size scale
    D. None of above

25. The true length of a vertically drawn edge of an object is 50 mm. What is the image length of the object given that the scale of the drawing is 1:2.
    A. 5 mm
    B. 10 mm
    C. 25 mm
    D. 50 mm
    E. 100 mm

$\frac{1}{2} = \frac{x}{50}$

26. Pick the odd one out.
    A. Parabola
    B. Cycloid
    C. Ellipse
    D. Hyperbola
    E. None of the above.

27. Circles, Ellipses, Cycloids, and Trochoids are all examples of:
    A. Loci
    B. Conic Curves
    C. Polygons
    D. Tangents
    E. None of the above

28. The following are properties of Orthographic projection except:
    A. The rays are orthogonal to the planes of projection,
    B. Only one side or surface is exposed to the rays,
    C. The rays are parallel to one another,
    D. It is a single-view projection,
    E. It is a multi-view projection

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INDEX NUMBER.....YEAR.....

9. In a reduction scale drawing, the drawn size is of:

A. The same size as the object
B. Reduced size of the object
C. Bigger size as the object
D. Only some area that is reduced
E. None of the above.

10. The two points that determine the accurate position of a line should be _____

A. Reasonably far apart,
B. Reasonable close to each other,
C. Either close or far apart
D. None

11. The following are all types of projection ***except***

A. Parallel projection
B. Perspective projection
C. Oblique projection
D. Axonometric projection
E. None of the above

12. In producing First Angle Orthographic Projections of an object, the object is be assumed to be placed in _____ quadrants respectively.

A. Transparent box system
B. Opaque box system
C. Second quadrant
D. Forth quadrant
E. None of the above

13. The first most important view of an object during Orthographic Projection is captured on:

A. Profile plane,
B. Horizontal plane,
C. Vertical plane
D. Auxiliary plane
E. Horizontal auxiliary plane

14. Figure 1 shows objects A-E. Select one of the objects from "B" to "E" that would fit exactly into object "A"

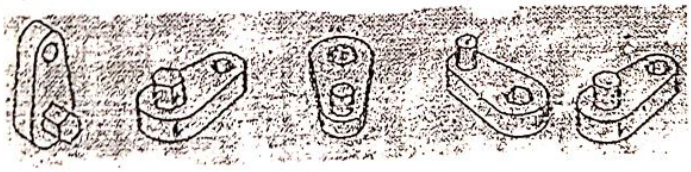


Fig. 1

A B C D E

15. The following types of projections can be categories as Axonometric Projection except:

A. Dimetric projection
B. Trimetric projection
C. Isometric projection
D. Perspective projection
E. None

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KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

B.Sc. (Engineering) First Semester Examination, 2013

First Year

ME 159 TECHNICAL DRAWING

DECEMBER, 2013

THREE HOURS

Index No:..... Department:.....

Answer All questions in Section A
All questions in Section B
One question in Section C

Section A

Provide appropriate responses to the questions in this section on the question paper and submit the question paper to the invigilator.

1. The scale of a drawing is the ratio ..... : .....

2. In a reduction scale drawing, the drawing is of:
   - A. The same size as the object
   - B. Reduced size of the object
   - C. Bigger size as the object
   - D. Only some area that is reduced
   - E. None of the above.

3. The image length of the edge of an object is 10 mm. What is the true length of the object given that the scale of the drawing is 1:5.
   - A. 2 mm
   - B. 10 mm
   - C. 25 mm
   - D. 50 mm
   - E. 50 cm

4. In the First angle projection, the object is placed in the ..... quadrant.

5. Draw a centre line in the space provided ( )

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36. Figure 2 shows a pictorial view of a block. Select a view from the views on the right side of the figure that correctly represents the Front View.

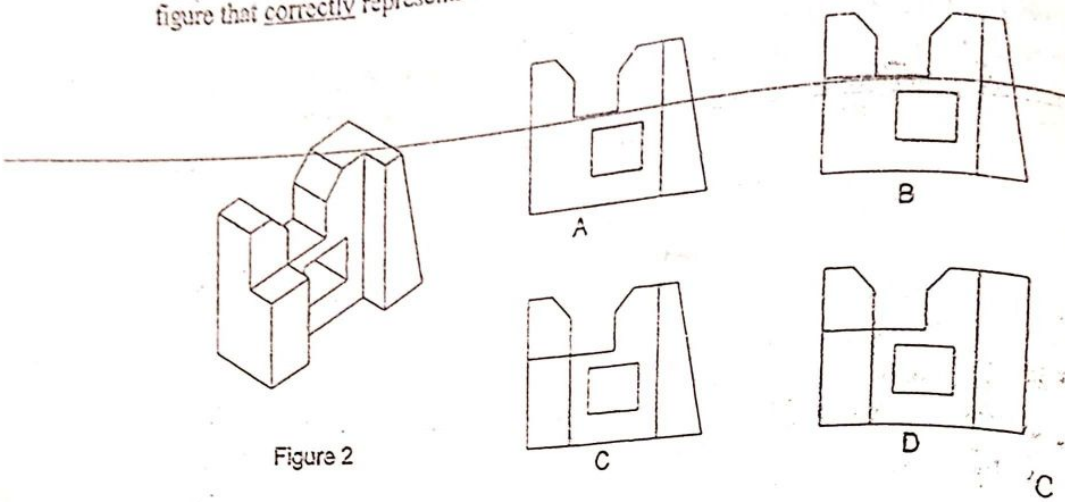


Figure 2

'C

Figure 3 shows a pictorial view of a block. Questions 37 - 40 are based on this figure.

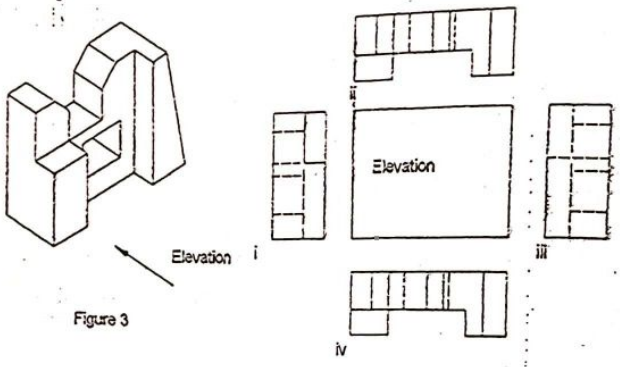


Figure 3

37. The view i. represents:
    - A. First angle orthographic projection Right Side View,
    - B. First angle orthographic projection Left Side View,
    - C. Third angle orthographic projection Right Side View,
    - D. Third angle orthographic projection Left Side View,

38. The view ii. represents:
    - A. First angle orthographic projection Top View,
    - B. First angle orthographic projection Bottom View,
    - C. Third angle orthographic projection Top View,
    - D. Third angle orthographic projection Bottom View,

39. The view iii. represents:
    - A. First angle orthographic projection Right Side View,
    - B. First angle orthographic projection Left Side View,
    - C. Third angle orthographic projection Right Side View,
    - D. Third angle orthographic projection Left Side View,

40. The view iv. represents:
    - A. First angle orthographic projection Top View,
    - B. First angle orthographic projection Bottom View,
    - C. Third angle orthographic projection Top View,
    - D. Third angle orthographic projection Bottom View,

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14. The following are properties of Orthographic projection except:
    A. The rays are orthogonal to the planes of projection,
    B. Only one side or surface is exposed to the rays,
    C. The rays are parallel to one another,
    D. It is a single-view projection,
    E. It is a multi-view projection

15. A student is requested to purchase clutch (mechanical) pencils for producing engineering drawings. Which combination pencil sizes will suit this purpose?
    A. 0.7 and 0.35
    B. 0.7 and 0.5
    C. 0.5 and 0.35
    D. 0.7 and 0.3
    E. 1.0 and 0.7

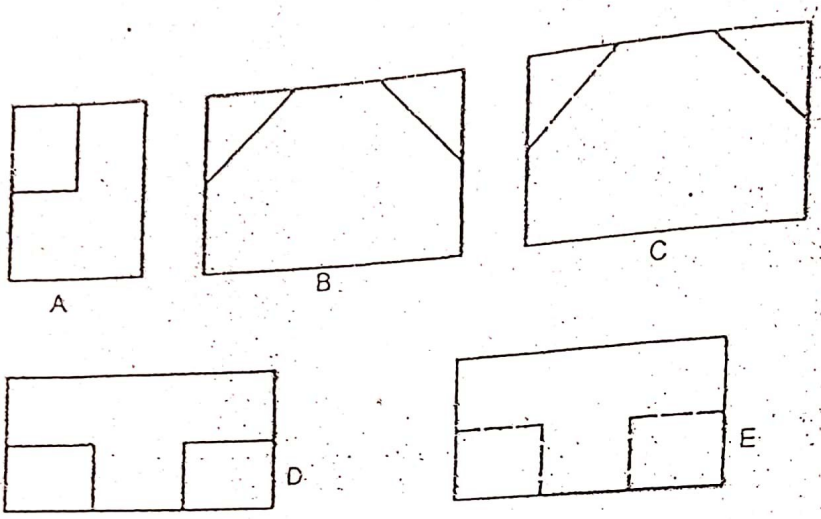
16. Hidden edges and outlines can be represented by:
    A. Thin continuous lines
    B. Thin chain lines
    C. Thin short dashes
    D. Thick continuous line
    E. Thick chain lines

17. The following information cannot be found in the title block:
    A. Size of object
    B. Title of drawing
    C. Scale
    D. Names and dates of drafter, checker, designer
    E. Name of firm.

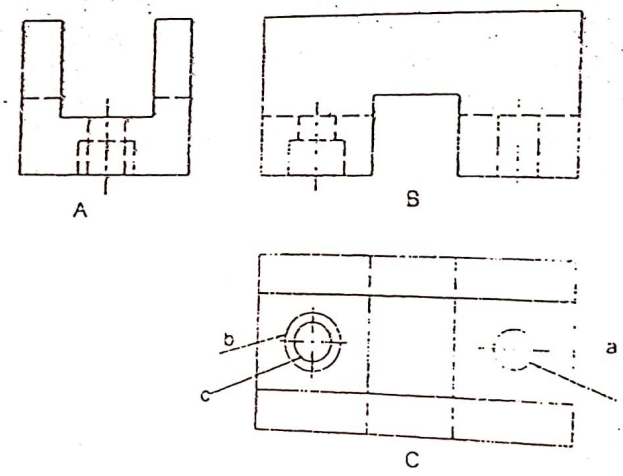
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18. The orthographic views of a 3-dimensional object are shown below. Which of the views will you select as the most important view (Front View)?



The figure below shows three views (A, B and C) of an object. Questions 19 and 20 are related to this figure.



19. For a First Angle Orthographic Projection, the view A is called:

A. Front View, B. Plan C. Right Side View, D. Left Side View E. None

20 Looking at View C, which of the features labelled (a, b and c) represents a hole?

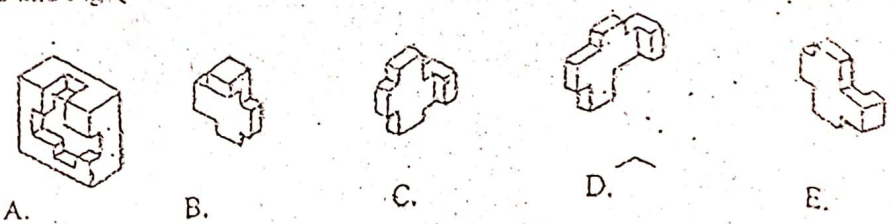
A. "c", B. "b" C. "c", D. "b" and "c" E. All

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In the Fig Q6 and Fig Q7 select one of the objects "B" to "E" that exactly fits into object "A"

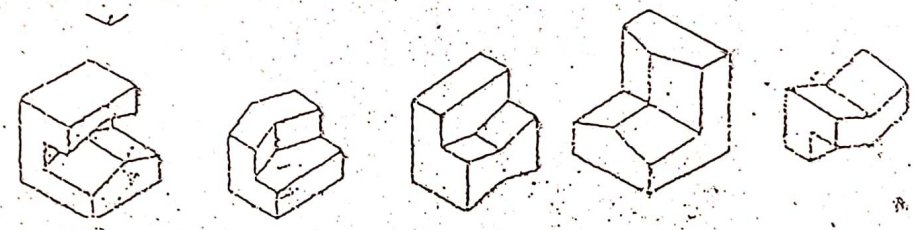
6.



A. B. C. D. E.

Fig Q6

7.



A. B. C. D. E.

Fig Q7

8. The first most important view of an object during Orthographic Projection is captured on
   - A. Vertical auxiliary plane
   - B. Horizontal plane
   - C. Vertical plane
   - D. Auxiliary plane
   - E. Horizontal auxiliary plane

9. In the third angle projection, the object is imagined to be placed
   - A. Below the HP and behind the VP
   - B. Above the HP and in front of the VP
   - C. Above the HP and behind the VP

10. A surface of an object appears in its true shape when it is ..... to the plane of the projection.

11. What is the difference between Superior and Inferior Trochoid:

    .....

12. To draw a horizontal line, the appropriate drawing tool is .....

13. Given the 2H, 2B 3H, 3B, HB lead type pencils, arrange them in order of hardness with the softer one first..... 3B, 2B, HB, 2H, 3H .....

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9H, 8H, 7H, 6H, 5H, 4H, 3H, 2H, H, F HB, B, 2B

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UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI
COLLEGE OF ENGINEERING

B.Sc. (Engineering) Supplementary Examination, 2011
First Year

ME 159 TECHNICAL DRAWING

AUGUST, 2011 THREE HOURS

Attempt ALL Questions

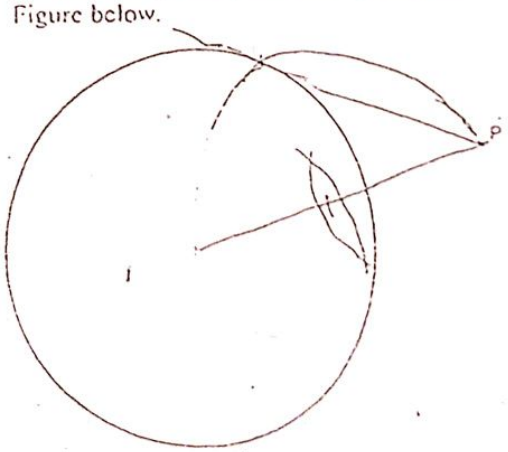
SECTION A (ANSWER ALL QUESTIONS ON THE ANSWER SHEET)

1. Hidden edges and outlines can be represented by:
   A. Thin continuous lines
   B. Thin chain lines
   C. Thin short dashes
   D. Thick continuous line
   E. Thick chain lines

2. The following information cannot be found in the title block:
   A. Size of object
   B. Title of drawing
   C. Scale
   D. Names and dates of drafter, checker, designer
   E. Name of firm.

3. The image length of the edge of an object is 10mm. What is the true length of the object given that the scale of the drawing is 1:5.
   A. 2mm
   B. 10mm
   C. 25mm
   D. 50mm
   E. 50cm

4. Construct a line that passes through the point P and tangential to the circle in the Figure below.



5. Determine the length of the arc shown in the figure below.

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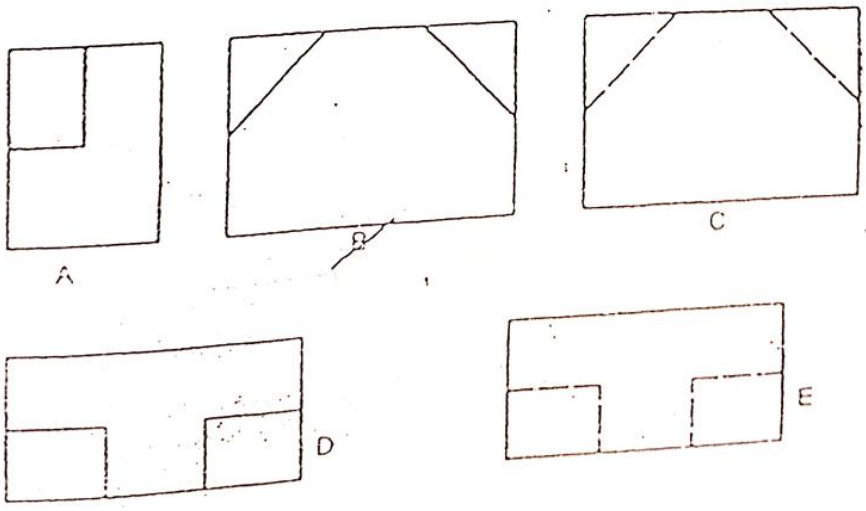
6. The first most important view of an object during Orthographic Projection is captured on:

   A. Vertical auxiliary plane,
   B. Horizontal plane,
   C. Vertical plane
   D. Auxiliary plane
   E. Horizontal auxiliary plane

7. The following are properties of Orthographic projection except:

   A. The rays are orthogonal to the planes of projection,
   B. Only one side or surface is exposed to the rays,
   C. The rays are parallel to one another,
   D. It is a single-view projection,
   E. It is a multi-view projection

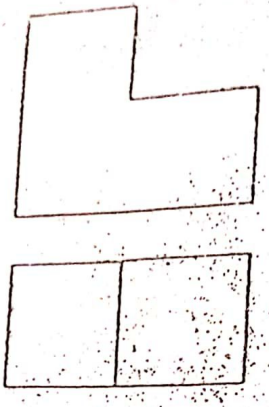
8. The orthographic views of a 3-dimensional object are shown below. Which of the views will you select as the most important view (Front View)?



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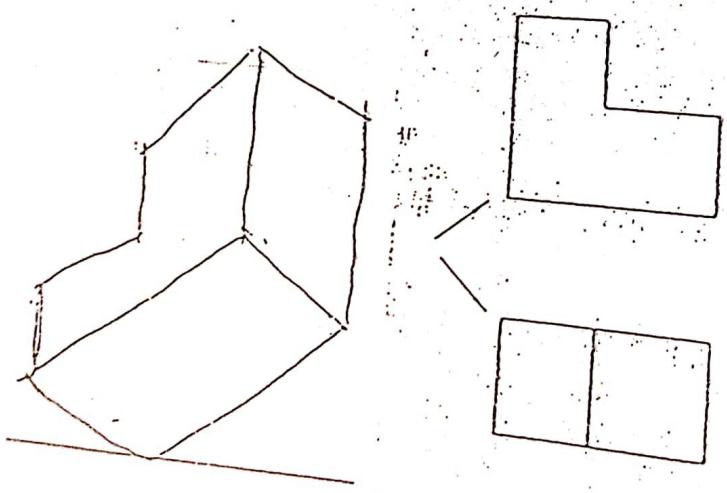
9. The 1st orthographic projection of a block is best and fully described by the two views shown below.



The best isometric projection of this same object will show the following surfaces:

A. Front-Top-Left
B. Front-Top-Right
C. Front-Bottom-Right
D. Front-Bottom-Left
E. None

10. Provide the isometric projection (shown by arrows) of the object whose first angle orthographic projection is shown below.



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Index No: .....

11. In the third angle projection, the object is imagined to be placed

A. Below the HP and behind the VP
B. Above the HP and in front of the VP
C. Above the HP and behind the VP

12. A surface of an object appears in its true shape when it is Parallel to the plane of the projection.

13. What is the difference between Superior and Inferior Trochoid:

Generating circle for Superior is bigger than that in Infe

14. To draw a horizontal line, the appropriate drawing tool is T-square

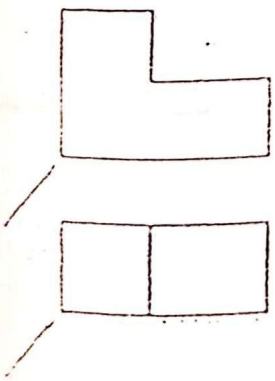
15. Given the 2H, 2B, 3H, 3B, HB lead type pencils, arrange them in order of hardness with the softer one first.

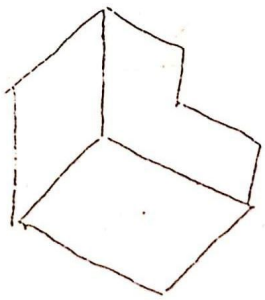
3B, 2B, HB, 2H, 3H

16. The following are properties of Orthographic projection except:

A. The rays are orthogonal to the planes of projection,
B. Only one side or surface is exposed to the rays,
C. The rays are parallel to one another,
D. It is a single-view projection,
E. It is a multi-view projection

17. The 1st Angle orthographic projection of a block is best and fully described by the two views shown in Fig Q17. Provide the isometric projection of this block based on the directions of the arrows shown in the figure:





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Index No.....

ME 159 SESSION

SECTION A

ANSWER ALL QUESTIONS

(30 marks)

1. Hidden edges and outlines can be represented by:
   A. Thin continuous lines
   B. Thin chain lines
   C. Thin short dashes
   D. Thick continuous line
   E. Thick chain lines

2. The following information cannot be found in the title block:
   A. Size of object
   B. Title of drawing
   C. Scale
   D. Names and dates of drafter, checker, designer
   E. Name of firm.

3 In first angle projection the object is placed in the .....4th.....quadrant

4 Divide the line AB in FigQ4 into seven equal parts

A _____ B

Fig Q4

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Index No.....

8 The first most important view of an object during Orthographic Projection is captured on:

A. Vertical auxiliary plane
B. Horizontal plane
C. Vertical plane
D. Auxiliary plane
E. Horizontal auxiliary plane

9 To draw a side view, an auxiliary vertical plane is imagined to be placed
A. Perpendicular to both the HP (Horizontal Plane) and VP (Vertical Plane)
B. Perpendicular to the HP and parallel to the VP
C. Perpendicular to the VP and Parallel to the HP
D. Parallel to the HP and parallel to the VP

10 The size of A4 sheet is 297 mm X 210 mm. What is the longest side of A2 sheet?
A. 420 mm
B. 594 mm
C. 840 mm
D. 1189 mm

11 The image length of the edge of an object is 10mm. What is the true length of the object given that the scale of the drawing is 1:5.

A. 2 mm
B. 10 mm
C. 20 mm
D. 25 mm
E. 50 mm

12 Draw a centre line in the space given ( )

13 In the third angle projection the object is imagined to be placed
A. Below the HP and behind the VP
B. Above the HP and in front of the VP
C. Above the HP and behind the VP.
D. Below the HP and in front of the VP

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Index No: .....

18. A student is requested to purchase clutch (mechanical) pencils for producing engineering drawings. Which combination sizes of pencils will suit this purpose?

A. 0.7 and 0.35
B. 0.7 and 0.5
C. 0.5 and 0.35
D. 0.7 and 0.3
E. 1.0 and 0.7

19. In a reduction scale drawing, the drawing is of:

A. The same size as the object
B. Reduced size of the object
C. Bigger size as the object
D. Only some area that is reduced
E. None of the above.

20. Construct a line parallel to line AB passing through the point D shown in the fig Q20.

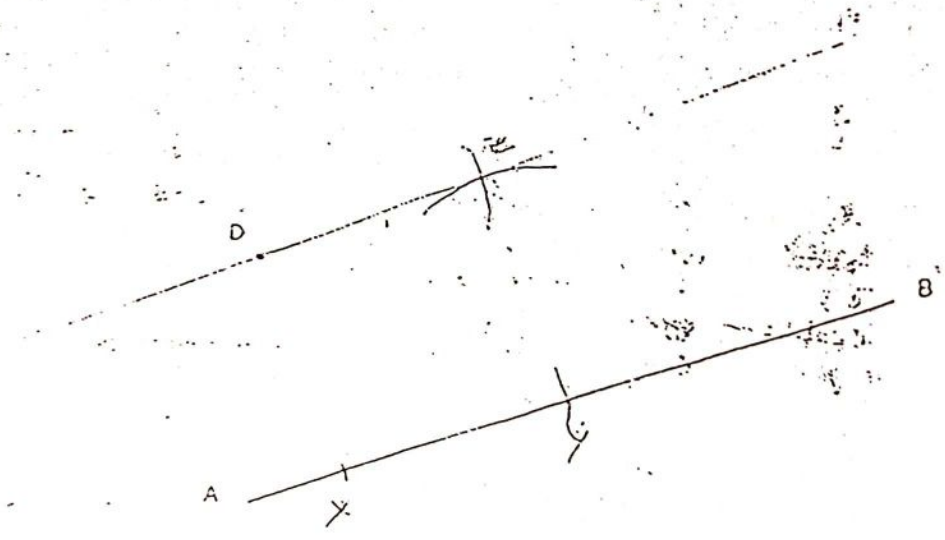


Fig Q20

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Index No.....

In the Fig Q18 and Fig Q19 select one of the objects "B" to "E" that exactly fits into object "A"

18

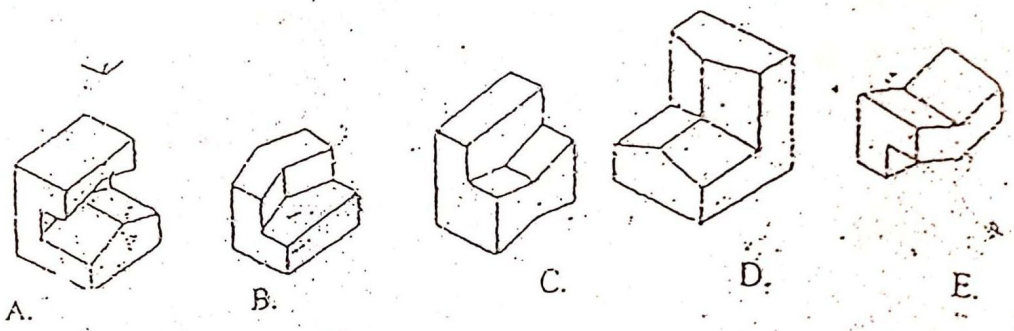
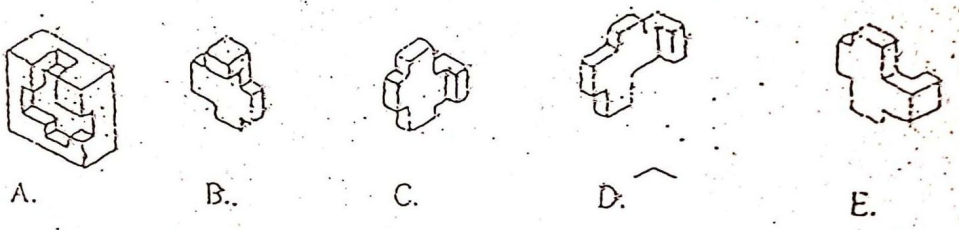


Fig Q18

19



20 A point P in space is projected on the vertical and horizontal planes as P_v and P_h respectively as shown in Fig Q20. Project the view of the point looking from the left side of the point

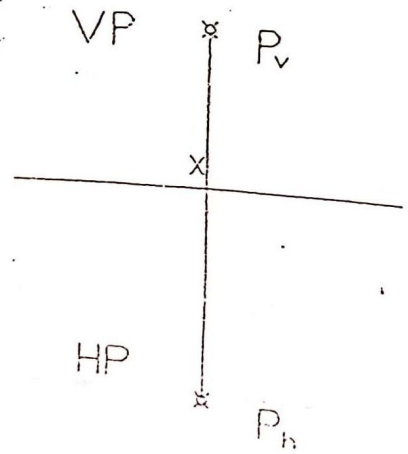


Fig Q20

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Index No.....

5 Construct the inscribed circle to the triangle ABC of Fig Q5

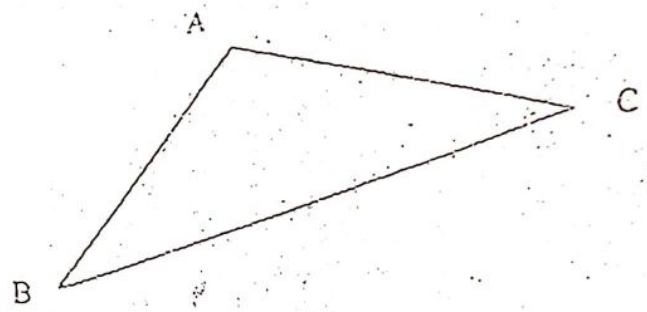


Fig Q5

6 Construct a line that passes through the point P and tangential to the circle in Fig Q6

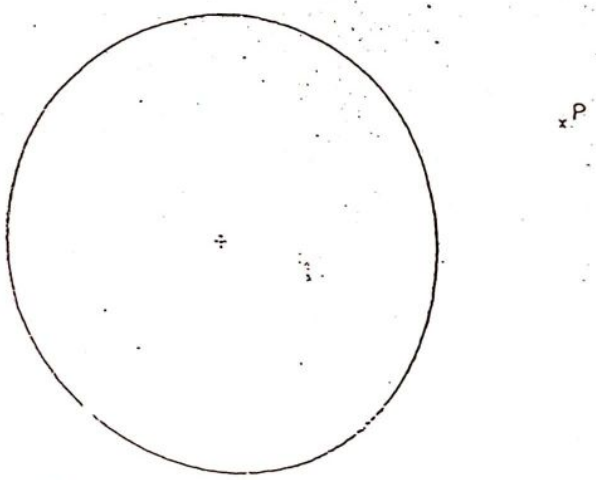


Fig Q6

7 In the first angle projection,

(a) The projection from above is placed .....the projection from the front

(b) The projection from the right is placed at the .....of the projection from the .....

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Index No.....

16 The major and minor axes of an ellipse are given in Fig Q16. Draw the ellipse using the approximate four arc or four centres method.

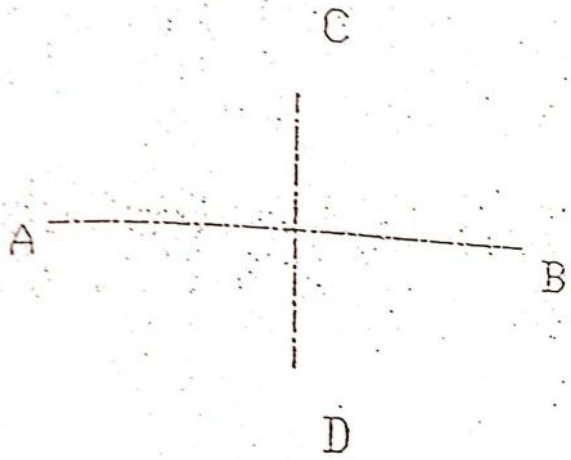


Fig Q16

17 An equilateral plane ABC, Fig Q17, has the height AH as 30mm. The triangle is now oriented such that the edge BC is perpendicular to the vertical pane and the triangle is parallel to the horizontal plane. Sketch as accurately as possible the three views of the triangle

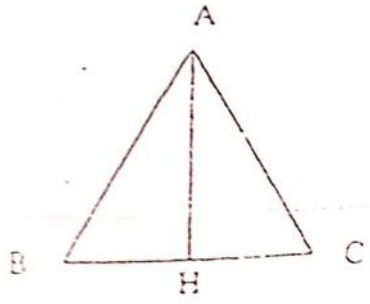


Fig Q17

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Index No.....

14 In the third angle projection, to obtain the view from left, the auxiliary vertical plane (AVP) is assumed to be placed on the .....of the object.

15. A rope is wound tightly on the square given in Fig Q15. Draw the involute of the square when it is unwound for three steps.



Fig Q15

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SECTION B
(20 marks)

Answer either Q1 or Q2 on the drawing sheet provided.

Q1 A wheel of diameter 80 mm rolls on a convex surface of radius 100 mm as shown in Fig Q1 B. Trace out the path of the point, P, located at a distance 45 mm from the centre of the rolling circle and at an angle of 30° to the vertical. Consider the curve for a quarter of a rotation. (20 Marks)

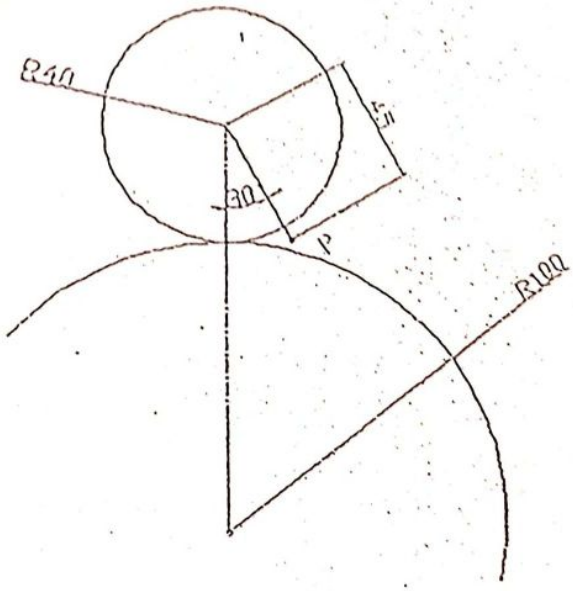


Fig Q1 B

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COLLEGE OF ENGINEERING

B.Sc. (Engineering) First Semester Examination, 2006
First Year

ME 159 TECHNICAL DRAWING

NOVEMBER, 2006 THREE HOURS

Attempt ALL Questions

Question 1. Figure 1 shows a bracket. By *purely* constructional means reproduce this bracket on your drawing sheet to a scale of 1:1.

28 Marks

Question 2. Figure 2 shows the first angle orthographic projections of a block. There is however a view missing. Provide the missing view (including the given two views) on the drawing sheet as well the isometric projection of the object whose three views you have produced.

14 Marks

Question 3 Figure 3 shows the isometric projection of a block. Provide the following first angle orthographic views:

i. Elevation, (Front)
ii. Plan, and
iii. Right-side

at a scale of 2:1. Do not add dimensions to the views.

28 Marks

J. Aveyire →
Afrifa – killer man
Ansong – Good man
Fiagbe / Wicked man
Sunnu
Davids

I dey feel what u rep.

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28 Marks

FELIX AGYEMANG

J. Aveyire →
Afrifa - killer man.
Ansong - God man
Fiagbe ✓ wicked man
Sunnu
Davids

I dey feel what u rep.

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Section B

Que. 1.a: Answer this question on the Question Paper.

i. Determine the centre of the arc in Figure .Q1a.

ii. Complete the arc to obtain a circle with centre C and by pure constructional means determine the circumference of the circle.



Figure Q1a

b) Re-construct Figure Q1b, on the Drawing Sheet, making all constructional lines visible.

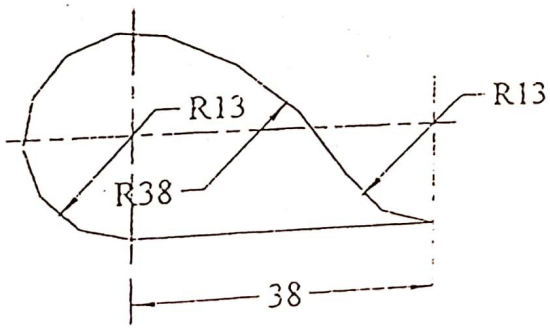


Figure Q1b

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2. A pictorial view of a bracket is shown in Figure Q2. Produce orthographic projections of the three principal views (state scale).

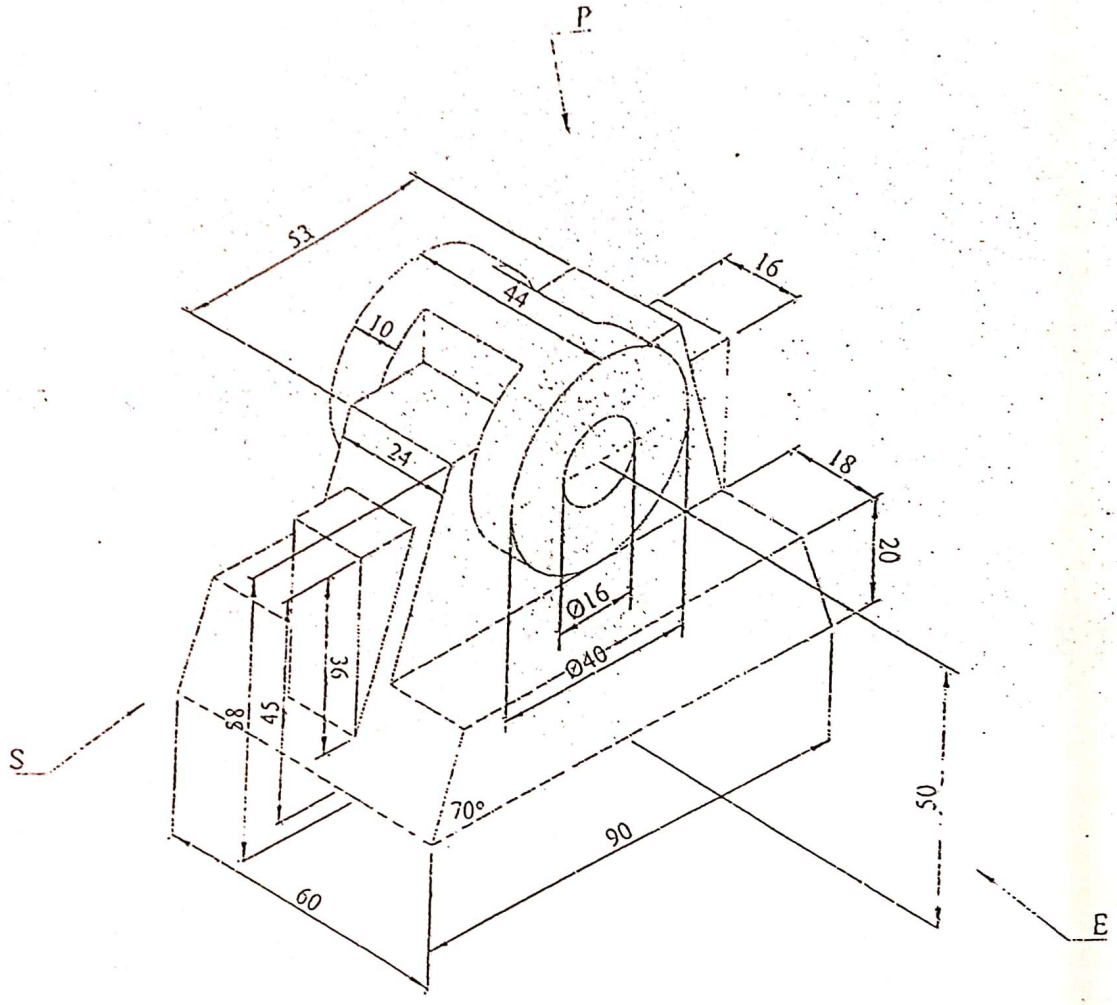


Figure Q2

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Q5 The front and plan views of an object are provided in Fig Q5C. Produce the isometric drawing of the object.

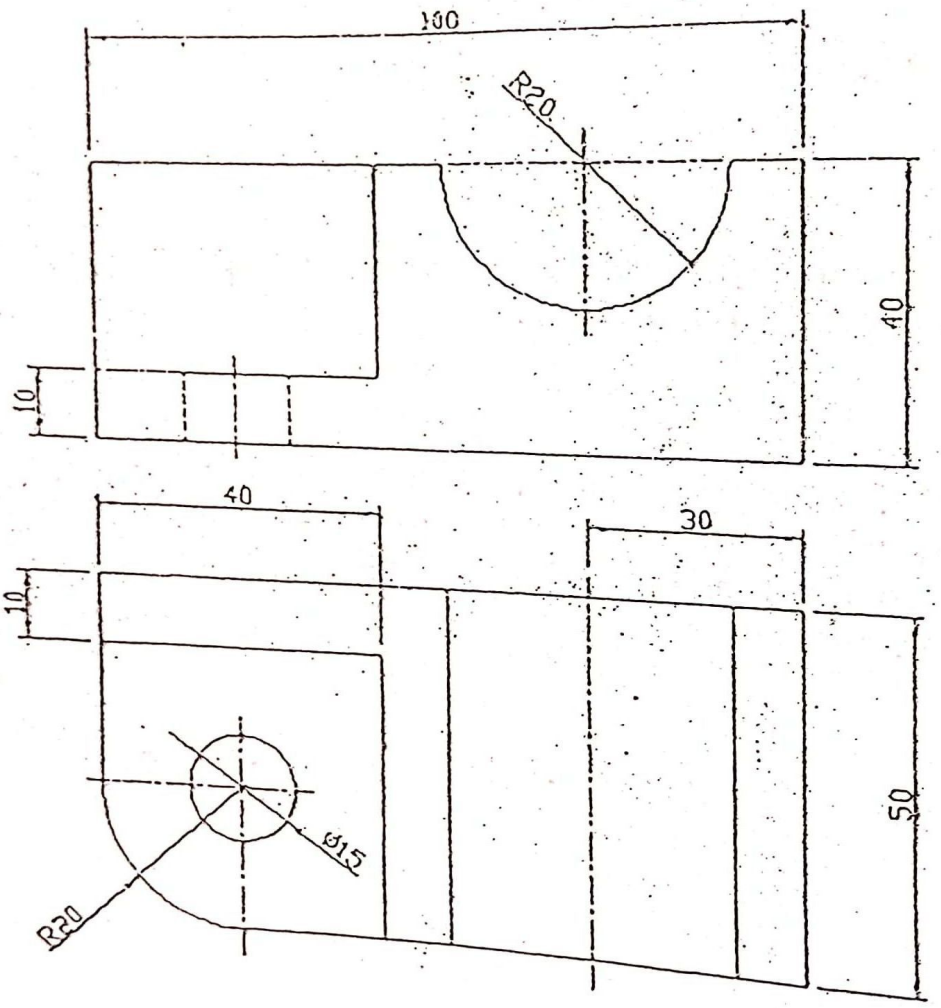


Fig Q5C

A.K. Sunnu
J. Aveyire
Y.A.K. Fiagbe
E.S.D. Afiti
C. Idan

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